

Analysis Plan:

Individual Water Insecurity Experiences (IWISE) Scale and Financial Well-Being

1 – BACKGROUND AND RESEARCH QUESTIONS

BACKGROUND

According to the WHO, 2.2 billion people did not have access to safely managed drinking water in 2022 (World Health Organization, 2023). These statistics are based on the WHO/UNICEF Joint Monitoring Program (JMP), which provides an infrastructure-based metric of water service access at a national level (World Health Organization, 2023). These metrics do not capture how people experience water insecurity and related stress in their daily lives. Water security is a multidimensional concept, spanning the affordability, reliability, adequacy and safety of water for well-being and a healthy life. Water insecurity can occur when at least one of these dimensions is not met (Jepson et al., 2017). The Water Insecurity Experiences (WISE) Scales can be used complementary to the JMP scale to measure water insecurity at the household and individual level (HWISE and IWISE, respectively) (Young et al., 2019a, 2021a).

Several studies using the WISE estimates have modelled low income as a predictor of water insecurity (Jepson et al., 2021; Mundo-Rosas et al., 2025; Stoler et al., 2020). Income alone, however, is not sufficient to assess financial well-being. Financial well-being is a multidimensional concept, combining objective dimensions such as financial assets, with subjective dimensions including personal perceptions of one's current financial situation and expectations for the future (Garg et al., 2024). Additionally, water insecurity and financial vulnerability could be mutually reinforcing: Evidence from low- and middle-income settings points to several plausible mechanisms such as affordability burdens, health burdens, productivity losses, and time- and labour-intensive coping strategies through which water insecurity could reinforce financial insecurity (Achore et al., 2020; Collins et al., 2024; Stoler et al., 2020). Yet, to our knowledge, no study directly estimates the association between water insecurity and financial well-being on an individual level.

RESEARCH QUESTION

The aim of this analysis is to estimate the association between individual water insecurity and financial well-being, and to identify factors that could influence this relationship. Individual water insecurity will be measured by the Individual Water Insecurity Experiences (IWISE) Scale. Financial well-being will be assessed using two complementary outcomes: an objective measure (Per Capita Income Quintiles), and an experiential measure (Financial Life Index, Gallup World Poll).

The overarching research question of this analysis is: What is the association between individual water insecurity experiences and financial well-being, and what factors influence this relationship?

Main analysis:

- To what extent is the IWISE scale associated with Per Capita Income Quintiles?
- To what extent is the IWISE scale associated with the Financial Life Index and its components?
- Which individual and contextual factors influence the association between IWISE and financial well-being?

Sensitivity and exploratory analyses:

- How do alternative cut-offs and water insecurity thresholds of the IWISE scale affect the association between IWISE scores and financial well-being?
- How does the relative contribution of different items of the IWISE scale affect estimates of individual water insecurity and how are they associated with financial well-being?

2 – MODELLING APPROACH

DATA AND VARIABLES

The main dataset used for this analysis is provided by Dr. Sera Young (Northwestern University, Evanston, IL) and was collected by the Gallup World Poll.

For the independent variable, the continuous IWISE score (0-36) is proposed as an indicator for experienced water insecurity, to keep as much information as possible and to facilitate modelling. The IWISE score is calculated based on 12 questions that evaluate the frequency in which individuals experienced water-related problems in the last 12 months. The items include questions on psychological well-being (worry, anger, shame), daily living (interruptions, changed plans, washing clothes), food and water intake (no water, limited drinking water, changing food, going to sleep thirsty) and hygiene (washing hands, washing body). The response options vary from 0-3 and are scored as “never” (0), “1 or 2 months” (1), “in some but not every month” (2) and “in almost every month (3), and are summed over all items. Individuals are usually classified as water insecure if their IWISE score is 12 or higher (Frongillo et al., 2024; Young et al., 2021b). If time allows, we are considering repeating the analysis with JMP-like measures from DHS and MICS surveys from the same countries to look for differences in the association with financial well-being between IWISE and JMP measures.

Two outcome variables will be extracted from the dataset to represent the multiple dimensions of financial well-being:

- Per Capita Income Quintiles
- Financial Life Index (FLI) by the Gallup World Poll

Per Capita Income Quintiles provide the objective indicator to assess financial well-being. They are derived from the Per Capita Annual Income in International Dollars and divide respondents into five income groups of equal size within each country. To capture the subjective dimension of financial well-being, the FLI will be used. It captures experiential aspects (similar to the IWISE scale) and includes four questions evaluating the lived financial situation of the respondent. For this analysis, a three-item version of the index including WP2319, WP30 and WP31 will be used, as the item WP88 is missing for 30% of respondents. The 3-item-index was calculated using the same methodology as the original index. The four-item FLI will be used for a robustness analysis. To see whether the association between IWISE and the FLI varies across its components, the items will each be modelled separately as well.

Individual and contextual factors potentially influencing the relationship between water insecurity and financial well-being are first identified by screening all the variables included in the Gallup World Poll dataset and later by screening global environmental databases. The inclusion of potential factors is based on conceptual relevance. Relevance is assessed via reasonable hypotheses and literature review, focusing on connections with water insecurity and/or financial well-being. In a next step, identified factors will be screened for redundancy and collinearity.

Covariates from the Gallup World Poll dataset

Table 1 Overview of included covariates from the Gallup World Poll dataset. Selection of covariates from the Gallup World Poll dataset is still ongoing.

Covariate	Reasoning
Gender	Research suggests that women often bear a disproportionate share of responsibility for water (Pouramin et al., 2020).
Urbanicity	In rural areas, drinking water services on premises are less common which influences water security (World Health Organization, 2021).
Age	Hypothesis: Likely to affect role within household and water-related burden.
Marital Status	Hypothesis: Likely to affect role within household and water-related burden.

Household size	Household size was found to be a mediating factor of water insecurity and food security on a household level (Shamah-Levy et al., 2025).
Employment	Employment status positively contributes to household water security (Asaki et al., 2024) and unemployment is associated with low income and poverty (Cai et al., 2021).
Education	Educational attainment has been connected to financial well-being (Assari et al., 2024) and is associated with better water access (Hargrove, 2020).
Food and Shelter Index	Water insecurity has been positively associated with food insecurity (Bethancourt et al., 2022), and there is a known association between financial standing and food insecurity (Smith et al., 2017).

Contextual covariates from global open access databases

Table 2 Overview of potential contextual covariates from global open access databases, reasons for inclusion, and data sources. The search for potential contextual covariates is not yet complete.

Covariate	Reasoning	Data source
Progress towards SDG 6.1	Hypothesis: The progress towards SDG 6.1 might cause a change in the perception of water insecurity (for cross-country comparison)	www.washdata.org
GDP (per capita)	Marker of economic development that captures country-level economic capacity, which affects both water insecurity and economic well-being (Nkiaka, 2022).	www.data.worldbank.org
Governance and Corruption	Good water governance has been linked with water insecurity (Ahsan et al., 2026; Miller et al., 2020) Indicator used from the WGI: Government effectiveness	www.worldbank.org
Diarrheal Disease DALYs	Diarrhoea accounts for the majority of the WASH-attributable disease burden with over 55 million DALYs in 2019 (Wolf et al., 2023).	ghdx.healthdata.org

METHODS AND STATISTIC

The data analysis will be conducted using the software Positron (2026.07.1). For data cleaning, the methodology from Young et al. (2022) will be followed. Within-country sampling weights, which were post-stratified by the Gallup World Poll to account for non-response and match the country's population characteristics, will be applied to the regression models. Projection weights created by Dr. Indira Bose will be used for cross-country comparison and for descriptive estimates. Where a country contributed to more than one survey wave, only a single wave was retained so that the same country is not counted twice in the pooled sample. Descriptive data analysis will be used to provide an overview of the dataset and its variables through summary tables and graphs.

The relationship between IWISE scores and both outcomes (Per Capita Income Quintiles and the FLI) will be assessed using multilevel ordinal logistic regression (the proportional odds model). The components of the FLI will each additionally be modelled separately, using multilevel binary logistic regression. The conducted pre-model checks and the results can be found in Table 3 . First, an unadjusted model including only IWISE scores and a random intercept for country will be fitted, with the IWISE score entered alongside its country mean to separate within- from between-country associations. The random intercept for country accounts for non-independence of respondents within countries. Second, a fully adjusted model will include all the covariates that were selected based on theoretical relevance. The model will be refit dropping one covariate at the time, to assess whether the IWISE coefficient changes substantively, defined as a change of 10% or more in the IWISE

coefficient. Once a minimum set of covariates is selected, interaction terms will be added individually to the adjusted model. To assess whether associations vary across settings, separate models will be fit for grouped data (such as regions and income-group levels) and results will be presented in forest plots, following the approach by Young et al. (2022). The analysis will be repeated using the 4-item IWISE scale, as well as the 4-item FLI, to check the robustness.

Ordinal logistic regressions estimate coefficients on the log-odds scale. These will be exponentiated and reported as odds ratios with 95% confidence intervals. These can be interpreted as the change in the odds of being in a higher outcome category per one-point increase in IWISE score.

A sensitivity analysis will be conducted in two parts. First, a cut-off analysis will assess how the water insecurity classification and estimated associations with financial well-being vary depending on whether a continuous scale, a single cut-off at 12 points (Young et al., 2019b, 2021b), or the four ordinal categories proposed by Frongillo et al. (2024) are used. The four ordinal categories identified by Frongillo et al. (2024) will be tested against alternative groupings and cut-off placements. Second, the individual items of the IWISE scale will be examined to assess their relative contribution to the overall IWISE score and estimated associations with financial well-being across different contexts.

Table 3 Results from pre-modelling checks

Preliminary check	Reasoning	Results
Missingness per variable	Listwise deletion is unbiased only if dropped rows are a random subset	- FLI sample: iwisescore 3.67%, log_gdp 1.25%, education 0.57%, all others <0.5%. - Income quintiles: iwisescore 3.54%, log_gdp 1.11%.
Dropped vs retained comparison	Tests the missing-completely-at-random assumption	- Not missing completely at random (MCAR). - Dropped cases had higher water insecurity in both models (FLI 8.45 vs 6.55; income 8.34 vs 6.19). - GDP near-identical (13,733 vs 13,812), indicating individual-level non-response rather than country coverage
Country coverage	Distinguishes module coverage gaps from item non-response	Four countries lost entirely: - EGY and MAR (FLI module not fielded) - VEN (100% missing in both models, no gdp data) - ZWE (INCOME_5 unavailable, as documented) - IND lost 74.8% from the FLI model only (approx. 9,460 rows, the largest single loss)
Three-item vs four-item responders for FLI	Levels 25 and 75 are reachable only from partial response	- WP88 not administered to 38,704 respondents. 52,462 answered four items, 27,342 answered three, 11,362 answered two (no index computed). - Confirmed as a fielding decision, not respondent refusal.
FLI recode validation	Confirms the reconstruction reproduces Gallup's published method	- Agreement = 1.00 across all 79,804 cases. - FLI_3item yields five uniformly constructed levels with no loss of sample.
Predictor distributions	Detects implausible values and severe skew	- iwisescore zero-inflated (median 3, mean 6.59, SD 8.42). - gdp_pc_ppp heavily right-skewed (mean 13,808 vs median 7,819), log-transformed. - hhsz maximum 30, plausible. - WGI_sc all 12–88, means near medians
Outcome × predictor cross-tabulations	Empty cells cause non-convergence and separation	No issues reported.
Events per variable	Requires approx. 10 cases per parameter in the smallest outcome category	- FLI_3item: N = 75070 parameters = 17 smallest category = 3587 EPV = 211.0 - Income_5: N = 85036 parameters = 17 smallest category = 13615 EPV = 800.9 (EPV = events per variables, rule-of-thumb: at least 10)
Correlation matrix	Detects pairwise redundancy among continuous predictors	- WGI severely collinear: ge/rq/rl/cc at $r = 0.90\text{--}0.94$; va/pv looser at $0.64\text{--}0.87$. - log_gdp × wgi_ge = 0.76. - iwisescore weakly correlated throughout (max $r = 0.28$)

VIF, full model	Detects three-way and higher overlap that pairwise correlation misses	No issues reported.
VIF, all six WGI indicators	Provides the evidence for retaining only one indicator of the governance covariates	Severe. wgi_rl 25.84, wgi_rq 13.23, wgi_cc 10.55, wgi_ge 10.00, wgi_va 5.00, wgi_pv 2.52. Four of six exceed the threshold of 10
Separation	Perfect prediction renders coefficients uninterpretable	No issues reported.

3 – REFERENCES

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